

Aman Obaid

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AI Backend Engineer building scalable APIs, event-driven systems, and production-grade infrastructure with a focus on performance, reliability, and developer ownership.

SKILLS

Language: Python, JavaScript, SQL

Backend: FastAPI, Flask, WebSockets

Frontend: HTML, CSS, Vue.js

Databases: PostgreSQL, Qdrant, MongoDB

Cloud & Infra: Redis, GCP, AWS, Docker

LLM Systems: LangChain, LangGraph, RAG

Observability: LangSmith, LangFuse, Git, Github

EXPERIENCE

Backend AI Engineer — Pinch Lifestyle Pvt Ltd

Aug 2024 – Jan 2026

- Built an **AI concierge assistant** serving **1000+** users personalizing conversations using profile preferences and conversation history, while automating tasks such as web search, flight and hotel bookings, and **CRM** service requests through **LangGraph**-based agent **orchestration**
- Built a production **RAG** pipeline over internal policy and operational PDFs using **LangChain** and **Qdrant**, enabling grounded responses from company knowledge and reducing hallucinations in concierge interactions.
- Built the **FastAPI** backend orchestrating **LangGraph** workflows, RAG retrieval, and tool execution, while asynchronously processing memory updates, preference learning, and post-response tasks using **Redis Queues** to keep API responses responsive
- Integrated flight booking, payment, and e-commerce services through an **MCP**-based tool layer with request validation, rate-limit handling, and recovery from partial failures for reliable task execution.
- Optimized system performance by implementing a **Redis**-based caching layer and refactoring **retrieval** to execute memory and **vector searches** concurrently, reducing redundant processing, lowering infrastructure costs, improving repeated-request response times by **45%**, and decreasing per-turn latency from **9.5s** to **5s**.
- Containerized and deployed **FastAPI**-based AI services on **Google Cloud** using **Docker**, implementing **CI/CD** pipelines and production monitoring to ensure reliable production deployments.

PROJECTS

GrabPic — AI Event Photo Discovery App

GitHub

- Built a biometric event photo search platform for galleries of **1000+ images** to enable instant personalized retrieval, using **FastAPI**, **RabbitMQ**, **Qdrant**, and **Redis** in a fully async processing pipeline.
- Designed a distributed facial embedding pipeline to sustain burst ingestion at **50 images/sec**, using queue-driven orchestration with **3-4 parallel workers**, idempotent processing, and retry-safe execution.
- Engineered **vector similarity search** over facial embeddings to return personalized results in **under 200ms**, with optimized **Qdrant** indexing across large collections.
- Secured biometric access with **JWT** authentication, face verification, duplicate prevention, and fault-tolerant indexing to protect user data end-to-end.

GoTrek — Role-Based Trek Booking Platform

GitHub

- Built **GoTrek**, a full-stack trek booking platform with a **Vue.js** client and **Flask** backend, implementing role-based authorization for admin, staff, and user portals across **25+ REST APIs** using **Flask-Security** and **SQLAlchemy**.
- Designed a **real-time** slot allocation and waitlist engine that prevented overbooking by automatically promoting the earliest waitlisted booking when cancellations freed capacity.
- Built an **asynchronous** task pipeline using **Celery** and **Redis** to decouple booking-history exports from the request cycle, running scheduled **Celery** Beat jobs for daily trek reminders and a monthly analytics report ranking the top 3 most-booked treks.
- Added a **Redis caching** layer on booking and trek-browse endpoints, using **5-minute** and **2-minute TTLs** with mutation-based invalidation to cut redundant database reads on repeat requests.

EDUCATION

Indian Institute of Technology Madras

B.S. in Data Science and Applications